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Permian Paleomagnetic Result from the Western Pyrenees Delineating the Plate Boundary between the Iberian Peninsula and Stable Europe

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The natural remanent magnetizations of Permian basaltic flows and Permian (?) dikes of the massifs of Cinco-Villas and La Rhune in the western Pyrenees have been studied. An unfolding test proves the characteristic magnetizations of the flows to be prefolding. From mainly thermal demagnetization analyses, a mean direction of magnetization has been calculated: $D = 217.5°$, $I = -11.3°$, $k = 25°$, and $\alpha_{95} = 7°$ for $N = 17$ sites (65 samples) and a virtual pole position at 40°N, 126°E. The characteristic directions of magnetization in the samples from the two dikes are conformable with those from the Permian flows. The paleodeclination of this result does not deviate in a counterclockwise sense from those from stable Europe, and it is concluded that this part of the Pyrenees was not rotated with the Iberian Peninsula. We have further delineated the fossil plate boundary between the Iberian Peninsula and stable Europe. The hypothesis of the evolution of the Bay of Biscay in two or more distinct stages is supported: the initial stage or stages imply a counterclockwise rotation of the Iberian Peninsula in Late Jurassic–Cretaceous times; later, a compressional Early Tertiary stage occurred during which the Iberian Peninsula moved northward with respect to stable Europe, thereby creating the Pyrenean orogen and its western continuation in the southern margin of the Bay of Biscay.

Analyses of geological and geophysical data from the Bay of Biscay and surrounding land areas have led many authors to accept hypotheses for the evolution of the Bay of Biscay within the framework of continental drift and plate tectonics (see recent volumes on the structural history of the Bay edited by *Debyser et al.* [1971]. Most common is the hypothesis that the Bay of Biscay was created during rotation of the Iberian Peninsula away from the European continental shelf off Brittany and southern England.

In this theory the initial position of the Iberian Peninsula is agreed upon by most authors as being approximately the position proposed by *Bullard et al.* [1965] in their fit of the Atlantic-bordering continents. Yet it is extremely difficult to reconcile all available geological and geophysical data within one simple plate movement of the Iberian Peninsula with respect to Europe. The proposals for this movement will be briefly reviewed.

One of the earlier proposed movements was a counterclockwise rotation of some 35° of the Iberian Peninsula around a pole of rotation in the Western Pyrenees [*Argand*, 1924; *Carey*, 1958; *Van der Voo*, 1969]. An objection to this movement was raised by *Mattauer* [1969] when he stated that this rotation also implies compression, or crustal shortening, over 35° in the Central and Eastern Pyrenees, for which there is little geologic evidence. Though large overthrusting and occasional nappe structures are known, they do not amount to the 100 to 200 km of required crustal shortening. Also it appears that there is an inconsistency in the timing: increasing evidence points to an age between Late Jurassic and Late Cretaceous for the creation of the Bay of Biscay, whereas the orogeny in the Pyrenees was during Early Tertiary, mainly Eocene, times.

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It was realized that a way to circumvent these difficulties was to have the plate movement take place along a curved fault or transform zone. This was considered but not elaborated on by *Van der Voo* [1969, p. 52]. Later *Le Pichon et al.*, [1971] evaluated the idea, presenting it in an attractive hypothesis implying rotation about a pole in northeastern France, which resulted in simple strike-slip movement along a curved fault tracing from the northern part of the Bay of Biscay, through the Aquitaine basin, to the North Pyrenean fault zone in the eastern Pyrenees (approximately a small circle around their rotation pole). They suggest that this movement was finished by Late Cretaceous times; in the Eocene the Iberian and European plates approached each other, thereby furnishing the compressional forces leading to the east-west trending Pyrenean orogeny.

Recently, with the advancing knowledge about the sea floor spreading episodes in the North Atlantic Ocean, *Pitman and Talwani* [1972] have also suggested that the movement of the Iberian Peninsula, with respect to Europe, may have consisted of two (or more) consecutive stages. First, in the Upper Jurassic and Lower Cretaceous the Bay of Biscay was created through coupled motion of the Iberian and the African plates away from the North American and European plates, thereby opening to some extent the Central Atlantic Ocean. This was perhaps followed by decoupling of the Iberian and African plates and further rotation of the Iberian Peninsula. In the Lower Tertiary a final stage occurred in which the African and the Iberian plate moved northward with respect to Europe, causing the Pyrenean orogeny. In this model as well, the earlier rotational movements of the Iberian plate cause only tensional forces in the Pyrenean realm, and one could expect that rifting and the formation of grabens would have led to the sedimentary basins in the Cretaceous, Paleocene, and Early Eocene.

The present paper provides a new paleomagnetic result from the Western Pyrenees, which, together with previously published data, enables a decision in favor of the recently proposed two-stage models.

Geological Situation of the Area and Sampling Details

The stratigraphic position of the Permian sequence of red sandstones and conglomerates with intercalated basaltic lavas is between Upper Carboniferous (Stephanian) fossiliferous shales and sandstones and Lower Triassic (Buntsandstein) pale red sandstones, conglomerates, and argillites [*Raguin*, 1954]. The latter sequence overlies the Permian with a slight unconformity. The Permian exposures are restricted to very few ocurrences; either the sequence was deposited in local pockets [*Raguin*, 1954] or, more likely, a larger area was covered and subsequently partly eroded. Up to four basaltic flows have been recognized in superposition in the three sampling areas (Figure 1). Their thicknesses varied between 10 and 40 meters. The entire Permian sequence has a thickness of up to 100 meters. It has not been metamorphosed; very few flows showed occasional slight alteration due to recent weathering.

Older Hercynian-folded Paleozoic sediments, a Late Carboniferous granite body, and the Permian and Lower Triassic sediments form the three tectonic massifs of the Western Pyrenees (Figure 1). The Permian and Lower Triassic Buntsandstein sediments generally dip away from the massifs, with dips up to 45° (Figure 1). The tectonic situation of the younger post-Buntsandstein sediments, on the other hand, is very complicated. They have been faulted, folded, and overthrusted in Early Tertiary times ('Pyrenean paroxism' of *Raguin* [1954]; 'compressional phase' of *Feuillée and Rat* [1971]). It is almost certain that the faulting and thrusting relates to major faults of the Hercynian basement: for instance, to the south of the massif of Cinco-Villas is an east-west trending zone of mylonitized rocks (Figure 1), including large chips of gneissic basement rocks that are nowhere else exposed. This zone appears to be the northern front of the 'nappe des marbres,' a mixture of tremolite-bearing marbles and unmetamorphosed sediments of the Jurassic and Cretaceous sequence, upthrusted toward the north. Associated with this mylonite zone are the so-called 'ophites' (Figure 1); gabbroic bodies or basic greenstones of questionable age, without contact metamorphism, tectonically emplaced and occasionally assum-

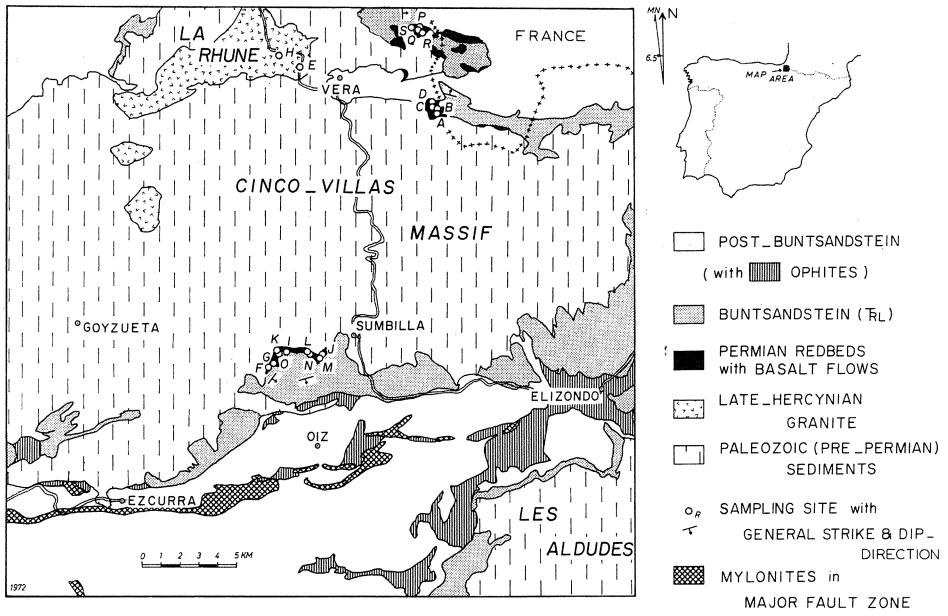


Fig. 1. Schematic geologic map of the area (Paleozoic massifs of Cinco-Villas, La Rhune, and Les Aldudes). The 17 sampling sites of the Permian flows and the two sites of the dikes are shown [after *Raguin*, 1954].

ing the role of lubricating layer in the same manner as the Keuper argillites and evaporites [*Raguin*, 1954; *Rutten*, 1969, p. 353]. We will consider the importance of these mylonite-ophite zones again in the last section.

Paleomagnetic sampling in this area took place in 1969 and 1971 at seventeen sites of Permian basaltic flows and two sites of Permian (?) dikes, intruded into the Late Carboniferous granite. Dips between 10° and 45° were measured in the sampled flows and the overlying sediments, the azimuths being to the south, the east, and the north (see Figure 1), thus permitting an unfolding test.

Analysis of the NRM

The measurements of the NRM with astatic magnetometers revealed intensities generally between 20×10^{-6} and 700×10^{-6} emu/cm^3 and Q values generally much higher than 1. Thermal and ac demagnetizations of 73 samples (17 sites) of the basaltic flows were carried out with standard methods employed at Utrecht University [*Zijderveld*, 1967*a*]. The initial measurements revealed a fair grouping of the directions of the NRM (Figure 2*a*).

The first demagnetization experiments were carried out with alternating fields. It appeared that about 80% of the samples had a very hard magnetization that was barely affected by ac fields up to 3000 oe (peak value), as can be seen in a typical example in Figure 3(*a*). This behavior indicated that for these samples thermal demagnetization was required. In Figure 3(*b*) the subsequent thermal demagnetization is presented for the same specimen as that of Figure 3(*a*). It is seen that 95% of the magnetization is eliminated in temperatures between 500° and 675°C, and that 50% of the magnetization is still present in the sample at 650°C, indicating that hematite is the carrier of the magnetization. Moreover, the vector of the magnetization does not change direction, suggesting that one component of magnetization is present in these samples. About 25% of the samples contained small secondary magnetizations; a typical example is given in Figure 3(*c*), where the secondary component is eliminated in temperatures below 550°C. The demagnetization diagrams of Figure 3(*e* and *f*) are exceptional rather than typical: in about 20% of the samples a considerable secondary magnetization was present. It was generally possible, however, to measure accurately the direction of the hardest (characteristic) magnetization eliminated between 650°C and 675°C, as can be seen in

Figure 3(*f*), which is an enlarged part of Figure 3(*e*). The results from eight samples (see Table 1) were discarded because of aberrant directions of magnetization.

A thermal demagnetization diagram of a sample from the Permian (?) dikes is given in Figure 3(*d*). Judging from the Curie temperature, magnetite appears to be the main carrier of the remanent magnetization; occasionally small secondary magnetizations are present.

The characteristic directions of magnetization, obtained from the demagnetization analysis, are presented before unfolding (Figure 2*b*) and after unfolding (Figure 2*c*). The unfolding test [*Graham*, 1949; *McElhinny*, 1964] is statistically significant on both the sample and the site level (Figure 2, Table 1), suggesting that the characteristic magnetizations are prefolding. The characteristic magnetization directions obtained from the samples of the two dikes in the granite body are conformable to those from the Permian flows (Figure 2*d*).

Giving unit weight to site mean directions of magnetization, a mean direction of magnetiza-

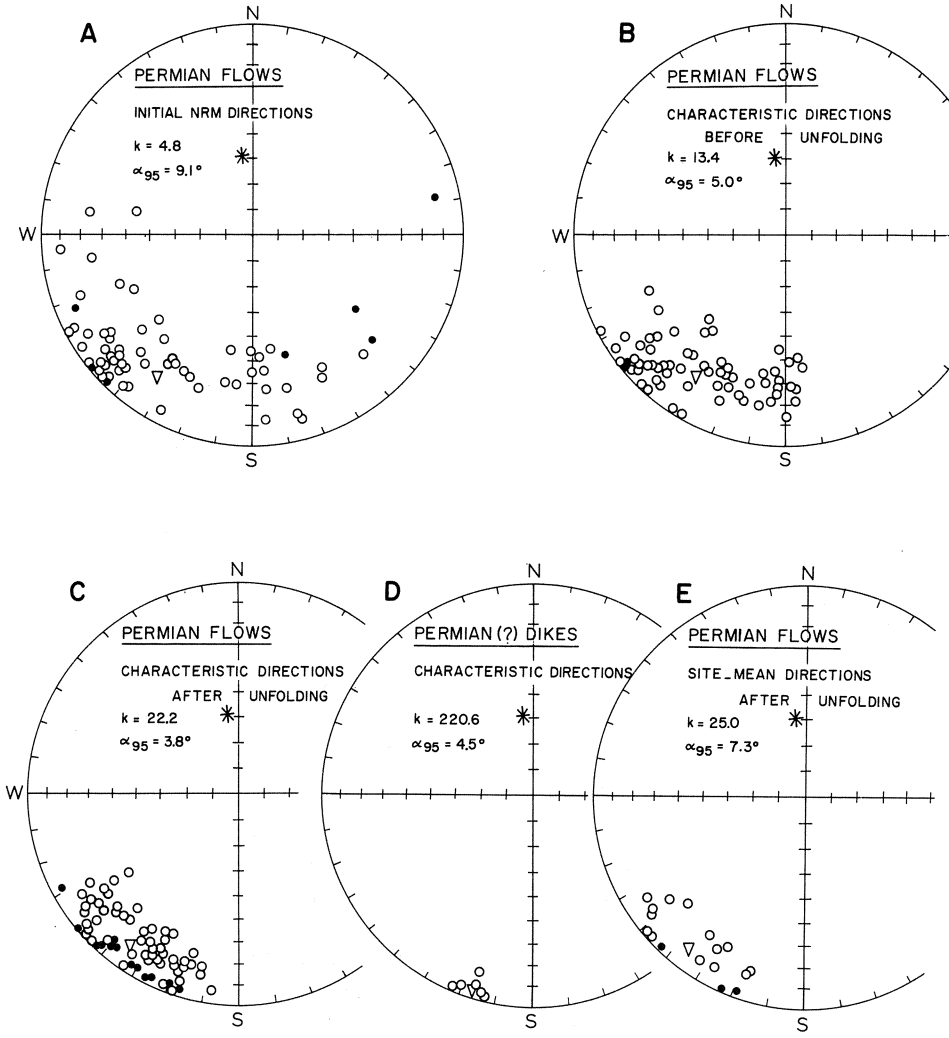


Fig. 2. Equal area projections of the directions of remanent magnetization of (*a*, *b*, *c*) the samples from the Permian flows, (*d*) the samples from the dikes, and (*e*) the site mean directions of the flows. Open symbols pointing upward, solid symbols pointing downward; the asterisk is the present-day local geomagnetic field direction; the triangles are the mean directions of magnetization, with the statistical parameters listed.

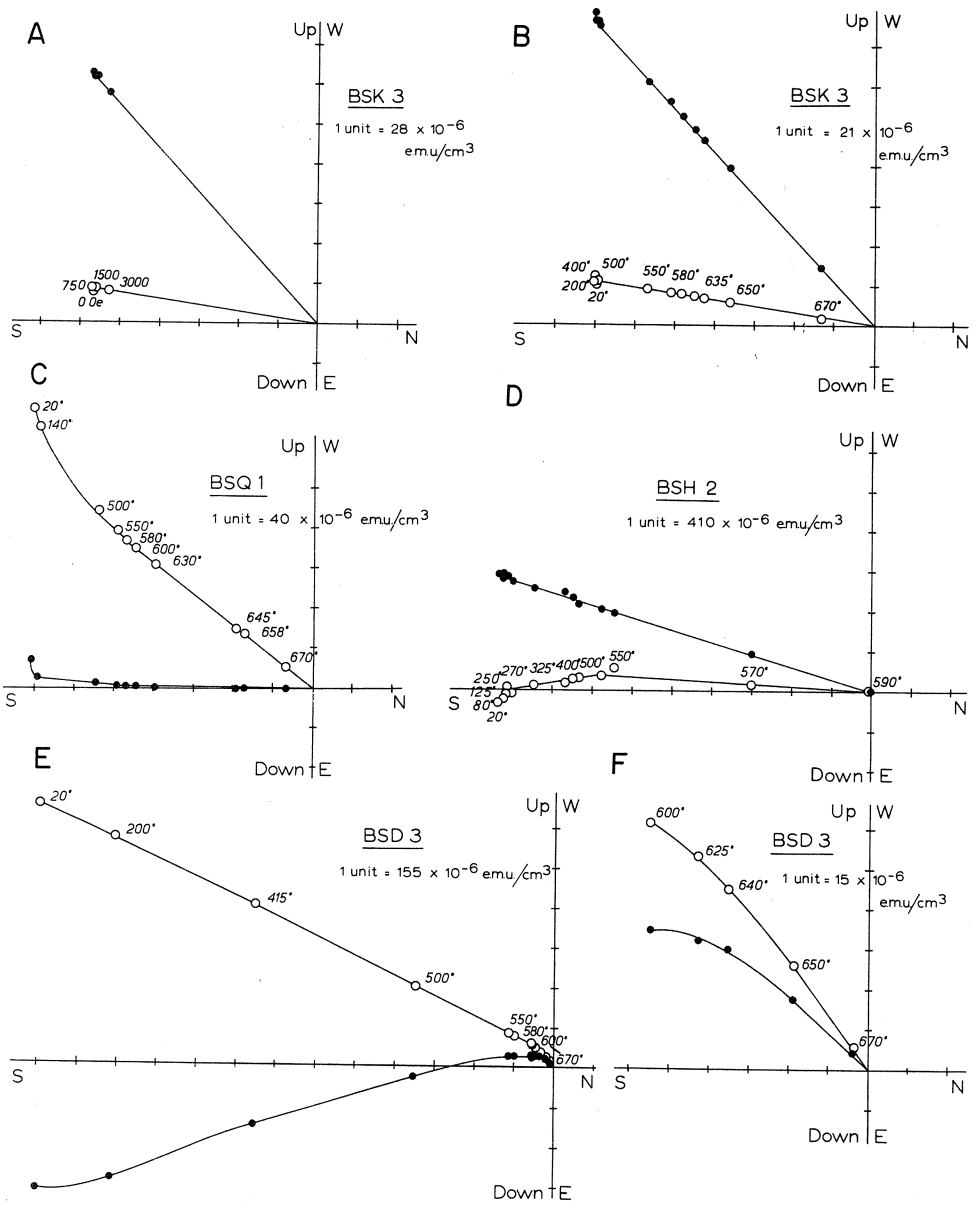


Fig. 3. Orthogonal projections of the successive end points of the magnetization vector during progressive demagnetization. Solid symbols represent projections on the horizontal plane, open symbols on the north-south vertical plane: (*a, b*) sample from site *K*; (*c*) sample from site *Q*; (*d*) sample from site *H* (dike); (*e, f*) sample from site *D*.

tion has been calculated having parameters $D = 217.5°$, $I = -11.3°$, $k = 25$, and $\alpha_{95} = 7°$ for $N = 17$ and a virtual pole position at 40°N, 126°E.

Discussion

The demagnetization analysis, combined with ore microscopy and electron microprobe studies, has shown that for most samples the magnetization is extremely 'hard' and resides in the fine-grained hematite particles that were observed. This points to a high oxidation state of the flows. In our opinion there is little doubt that the hematite is a secondary phase, but most likely of Permian age, as we will try to illustrate.

In the samples microscopically examined with

TABLE 1. Summary of NRM and Site Mean Directions of Characteristic Magnetization of the Permian Basaltic Flows of the Western Pyrenees Obtained by Progressive Thermal Demagnetization

						Characteristic Directions of Magnetization					
		Total NRM				Before TC		After TC			
Site	R_s	D	I	α_{95}	k	D	I	D	I	α_{95}	k
A	2/3	208.5	-29.5	7.9°	999	203.0	-29.8	199.6	+2.6	9.5°	688
B	4/6	209.5	-14.8	58.0°	4	208.2	-30.4	203.8	+0.9	5.1°	324
C	4/4	234.5	-12.8	59.5°	4	232.8	-25.0	225.9	-2.5	15.9°	34
D	5/6	216.8	-30.1	26.8°	9	222.6	-36.4	213.0	-8.9	10.1°	58
F	3/3	222.8	-7.3	10.1°	148	223.0	-7.7	223.9	+2.1	9.0°	190
G	1/1	203.5	-27.0			203.0	-29.0	215.1	-21.9		
I	1/1	224.5	+1.0			232.0	+1.0	233.3	-9.3		
J	1/1	232.0	+1.0			232.0	-0.0	233.6	-10.3		
K	6/6	229.5	-5.0	7.4°	82	229.1	-3.8	229.6	-1.1	7.3°	86
L	5/5	225.9	-19.7	8.5°	82	224.4	-20.3	227.8	-26.4	7.7°	99
M	1/1	236.5	-4.0			236.8	-4.8	237.7	-13.1		
N	6/6	231.3	-12.2	5.5°	147	231.5	-12.9	233.2	-21.2	5.2°	165
O	5/6	197.7	-28.9	21.2°	14	199.7	-23.8	209.8	-18.6	5.5°	195
P	5/6	187.9	-20.2	24.5°	11	180.4	-27.6	198.1	-14.6	5.5°	190
Q	5/6	179.5	-42.1	6.2°	149	181.8	-38.5	207.4	-21.1	6.8°	129
R	6/6	255.2	-17.8	54.9°	2	206.5	-15.3	208.3	-9.7	13.1°	27
S	5/6	131.0	-9.8	40.2°	5	181.6	-26.7	198.2	-13.2	8.6°	80
All samples:											
	65/73	213.5	-20.7	9.1°	5	211.7	-22.6	215.4	-11.9	3.8°	22
All sites:											
	17/17	215.6	-18.2	13.4°	8	214.5	-20.5	217.5	-11.3	7.2°	25

R_s is the ratio of the number of samples used for computation to the number of samples measured; D and I are declination and inclination in degrees; and TC is tectonic correction.

transmitted and reflected light, or analyzed with the electron microprobe, rutile, hematite, and a Fe-Ti-rich phase (probably ilmenite) were observed. It is inferred that observed two-phase grains, now consisting of ilmenite and fine-grained hematite, were produced by selective oxidation of magnetite in an original ilmenite-magnetite exsolution intergrowth. Cubo-octahedral outlines of these grains, coupled with the presence of ilmenite, suggest that the primary magmatic phase was a Ti-rich magnetite. This original phase was evidently accompanied by an Fe-poor ilmenite which now appears as platy crystals of ilmenite; these show secondary alteration to a transparent phase of high refractive index (possibly leucoxene), but they lack the abundant hematite characteristic of the cubic grains. There is little question that the development of the sooty hematite after magnetite is secondary, but it is probably deuteric in that a few minute grains of some copper-iron sulfide (probably chalcopyrite) appear within the hematite and do not themselves show signs of weathering. In any event, the secondary event is very probably Permian in age. This is most strongly supported by the fact that, when one compares the paleolatitude derived from the paleomagnetic direction of these flows with the Late Permian paleolatitude pattern obtained from Europe and North America, there is excellent agreement (Figure 4). The paleomagnetic poles from the periods in the Mesozoic or Tertiary are all significantly different from the Permian poles, the paleolatitudes extrapolated to the western Pyrenees always being much higher than the Permian values, which vary between the equator and 10°N [*Van der Voo and French*, 1973].

The mean paleodeclination is southwesterly, and it is of interest to compare it with other Permian data from western Europe and the Iberian Peninsula. In previous research, comparisons with Permian, Triassic, and Jurassic paleomagnetic data from the Pyrenees and from the Iberian Meseta have led to the conclusion that there are significant deviations between the

declinations from rocks of the stable part of Europe and those from the Iberian Peninsula south of the North-Pyrenean fault. Without generalization it can be noted that the paleodeclinations from stable Europe for these periods are SSW-NNE, whereas those from the Pyrenees and the Meseta are always SSE-NNW. This consistent deviation in declination has been interpreted as being due to a 35° counterclockwise rotation of the Iberian Peninsula [e.g., *Van der Voo*, 1969]. It is remarkable that the paleodeclination of the Permian flows in the western Pyrenees does not deviate in a counterclockwise sense from those from stable Europe, and we therefore conclude that this part of the Pyrenees was not rotated with the Iberian Peninsula. Consequently, the plate boundary between the Iberian Peninsula and stable Europe can be drawn with reasonable confidence as an east-west trending zone. In Figure 5 an attempt has been made to delineate this plate boundary as a major Pyrenean fault zone, following the en echelon arrangement of well-known faults. The tectonic features that accompany this major Pyrenean fault are, from east to west, as labeled in Figure 5:

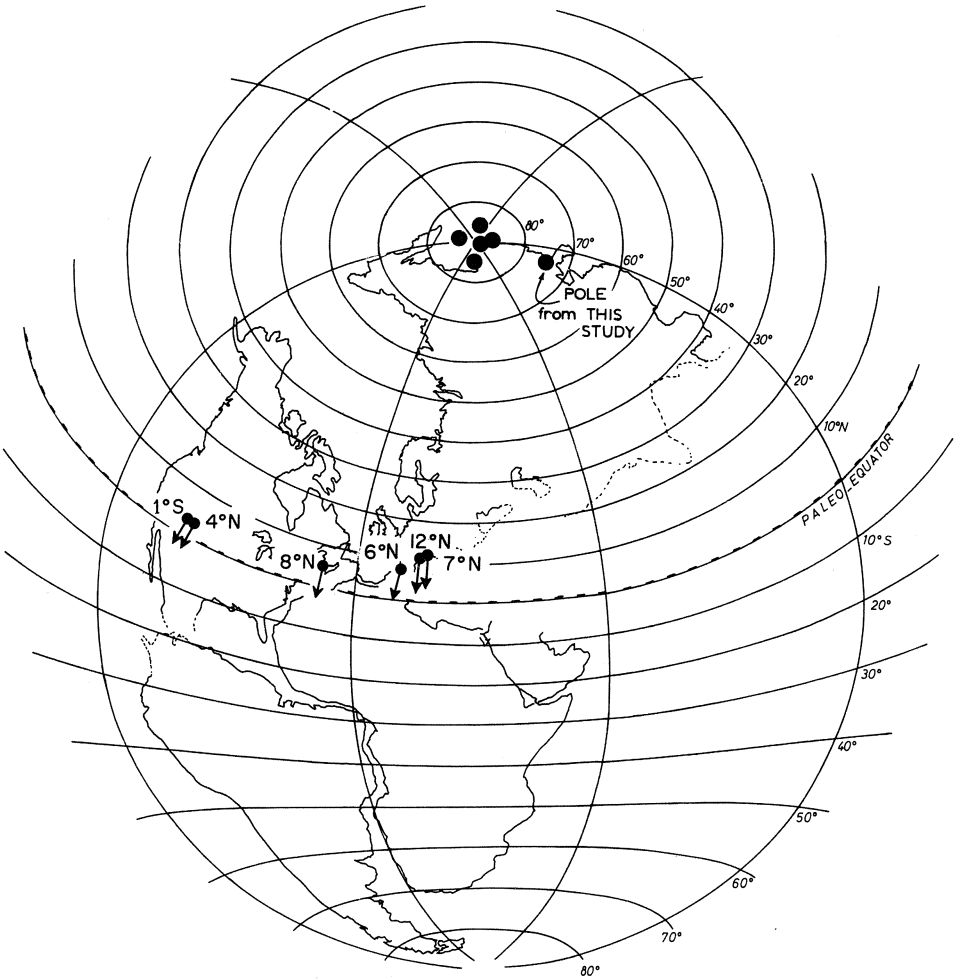


Fig. 4. Equidistant projection of the Atlantic-bordering continents in the reconstruction by *Bullard et al.* [1965]. The reliable Upper Permian paleomagnetic poles from stable Europe and North America have been plotted as dots; the directions of the declinations (parallel to the paleomeridians) and the paleolatitude values derived from these data are given at their sampling sites and can be compared with the paleolatitude pattern constructed from the mean Upper Permian pole position. Data are from *Farrell and May* [1969], *Larochelle* [1967], *Van den Ende* [1970], *Zijderveld* [1967*b*], and the present paper.

(*a*) Outcrops of ultrabasic rocks of lherzolitic type along the North-Pyrenean Fault, suggesting a deep-rooted suture that probably affected the entire continental crust [*Le Pichon et al.*, 1971].

(*b*) En echelon high-angle faults crossing the buried axial zone of the Pyrenees.

(*c*) The mylonite-ophite zones, mentioned before, to the south of the paleozoic massifs of the western Pyrenees, separating different sedimentary facies on either side [*Raguin*, 1954; *Feuillée and Sigal*, 1965]; again the ultrabasic rocks suggest deep-rooted phenomena.

(*d*) Late Cretaceous effusive rocks (pillow lavas) in the synclinorium east of Bilbao. Otherwise there is no indication of a major fault here.

(*e*) To the west the tectonic suture is submerged and may continue along the continental margin off northern Spain, recognized to be the western extension of the Pyrenean orogen.

Several sedimentary features accompany this line in the western Pyrenees and Cantabria, expertly described by *Feuillée and Rat* [1971].

This major east-west tectonic lineament of Figure 5 appears to be the most likely surface expression of the fossil plate boundary between the Iberian Peninsula and stable Europe, deformed by the Pyrenean orogeny into a 'suture.' Its delineation differs in the western Pyrenees from the fault trace proposed by *Le Pichon et al.* [1971], though their model can be modified by introducing a combination of spreading centers and transform faults in the Pyrenean realm (Figure 6*b*). The pole of rotation of *Le Pichon et al.* [1971], however, is not the only possible one and, in fact, one could argue that there probably have been 'infinitesimal' rotations around a sequential series of poles of rotation. We prefer a more general model which leaves room for these possibilities in a hypothesis for the evolution of the Bay of Biscay. Accordingly, we propose the following sequence of events: first, tensional stages during which the Iberian Peninsula rotated counterclockwise in Late Jurassic/Cretaceous times, around pole(s) of rotation that as yet remain ill-defined; later, a compressional stage during which the Iberian Peninsula moved northward

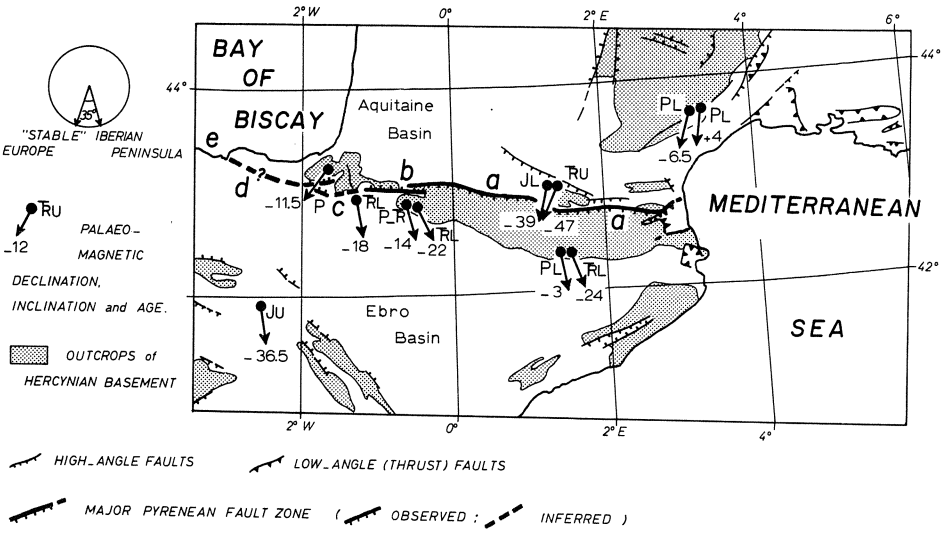


Fig. 5. Schematic tectonic map of the Pyrenees and surrounding areas. Paleomagnetic directions from Permian, Triassic, and Jurassic formations are plotted. The directions of the arrows, plotted where the rocks were collected, represent the declinations; the inclination values and the ages are indicated. Note the difference between the SW-SSW ('European') declinations and the SSE-SE ('Iberian') declinations. These paleomagnetic data are from *Girdler* [1968], *Kruseman* [1962], *Schwarz* [1963], *Stauffer and Tarling* [1971], *Van der Lingen* [1960], *Van der Voo* [1969], *Van Dongen* [1967], and the present paper. A major Pyrenean fault zone as the plate boundary between the Iberian Peninsula and stable Europe is proposed; the letters *a*, *b*, *c*, *d*, and *e* are explained in the text.

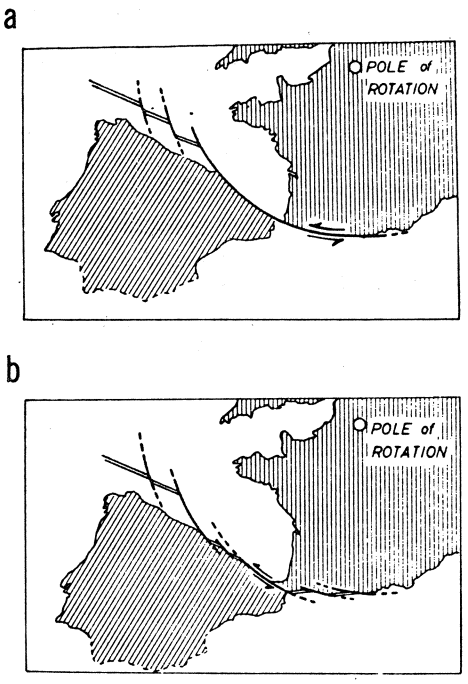


Fig. 6. Schematic positions of the plate boundaries between the Iberian Peninsula and stable Europe: *(a)* after *Le Pichon et al.* [1971, Figure 14]; *(b)* modification of *a* as required by paleomagnetic data.

or northwestward, thereby creating the Pyrenean orogen and its western continuation in the southern margin of the Bay of Biscay.

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